TRIVIALITY OF THE STRUCTURE SHEAF PUSHFORWARD FOR FAMILIES OF GEOMETRICALLY REDUCED CONNECTED FIBRES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The pushforward f_*O_X equals O_S and is preserved by base change once the natural map kappa(s) -> H^0(X_s, O_{X_s}) is an isomorphism for all s. invoked at proof lines 8325-8327. Reduces the statement to the fibre computation.
- The global sections of a proper geometrically reduced scheme over a field are a geometric field extension, and geometric connectedness forces the constant functions. invoked at proof lines 8328-8330. Gives H^0(X_s, O_{X_s}) = kappa(s).

Source reference. Original lemma 0E62: sources/stacks-perfect/pinned-original-source.tex, source lines 8264–8264. Cited at proof lines 8325-8325 for the reduction to the fibre computation; the fibre computation is inside the counted span.

Source reference. Original lemma 0BUG: sources/stacks-chow/tags.txt, source lines 14874–14874. Cited at proof lines 8328-8329 for the fibre computation; the conclusion is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Assume*

(1) *f is proper, flat, and of finite presentation, and*
(2) *the geometric fibres of f are reduced and connected.*

Then $f_*\mathcal{O}_X = \mathcal{O}_S$ and this holds after any base change.

Proof. By Lemma 0E62 it suffices to show that $\kappa(s) = H^0(X_s, \mathcal{O}_{X_s})$ for all $s \in S$. This follows from Varieties, Lemma 0BUG and the fact that X_s is geometrically connected and geometrically reduced. □

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